

Dr. Anthony Brew

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Customer-Focused AI executive, researcher, and engineer. 20+ years in AI since 2003, across academia, startups, scale-ups, and enterprises. Remains hands-on across the AI and engineering stacks while leading principal-level ICs and managers; scales expert units (<10) to multi-department orgs (>50), shipping state-of-the-art AI products that minimise technical uncertainty alongside continuous customer impact.

Overview

Highlights: Ph.D. Machine Learning (16 publications) · M.Sc High Performance Computing · Inventor on 16 patents · University Mathematics Tutor (Trinity College Dublin) · Zalando SE Supervisory Board (2018-2020) · Stanford XCS224N – Natural Language Processing with Deep Learning (2020)

Hard Skills: Generative AI, LLMs & Agentic Systems, Model Context Protocol (MCP), AI Gateway Orchestration, Recommender Systems & Contextual Bandits, AI Content Authenticity, Automated Content Moderation, Audience Targeting, Text and Image Processing, Classical Machine Learning & Deep Learning, Anomaly Detection/Statistical Process Control, Software Engineering & Distributed Systems

Soft Skills: Research Management, Product Management, Cross-Functional Team Leadership, Executive Stakeholder Engagement, Systems Design, Strategic Planning, Program Management, Agile Methodologies

Recent Technologies: OpenAI, Google Gemini, MCP, Claude Code, Codex, Replicate, PyTorch, Snowflake, Spark, Kafka, MLflow, AWS (Sagemaker, Lambda), Terraform, GitHub Actions

Experience

Senior Director of AI :: Shutterstock :: Dublin, Ireland :: *July 2023 - Present*

Reports to CTO. Org of 27 across 3 product teams (Generative Experience, Public API, Optimisation) plus a Principal Engineering group (peak 40).

Shutterstock is a two-sided stock-media marketplace; Generative AI both disrupts and extends the business model.

Image generator at 70%+ monthly retention; fastest-growing media on Shutterstock. I own technical strategy for safe, indemnified GenAI on vendor models (OpenAI, Google, Black Forest Labs, ByteDance, and others), with foundational models previously co-built with Databricks and Nvidia. Three product pillars:

- **Generative Model Optimisation & Recommendation:** LinUCB contextual bandit picks the best generator from user prompt and page context and learns from clicks. +20% over baseline, +30% on vendor models. Purpose-built traffic-routing doubled scientist deployment velocity.
- **AI Gateway, Public API & MCP – the orchestration spine:** My teams build the unified AI Gateway and Public API/MCP – the foundational layer for AI across Shutterstock. Routes all third-party calls (OpenAI, Google, others) under one governance layer; runs every AI PR review. Our API-first approach exposes Recommendations and Safety as agent-addressable, powering the Shutterstock Generative Experience, the official Shutterstock app in ChatGPT, and chat-based search – ready for broader consumption.
- **Safety, Indemnity & Content Authenticity:** Moderation underpins customer indemnity on commercial use of generated assets. Decision automation: 0% → ~60% of Image assets; Video rollout underway. Brand-safety pillars (e.g. child safety) hold ~100% recall at production precision. State-of-the-art AI-content detector distinguishes camera-captured from algorithm-generated content – load-bearing for indemnity decisions and customer trust signals.

Beyond product, reshaping engineering for the AI-coding era:

- **AI Community of Practice – Agentic Engineering Adoption:** Drove a company-wide AI Community of Practice rollout. Seeded with the "AI(All)ready" early-adopter cohort, scaled via peer mentorship and automated coding standards, culminated in a hackathon. 75% voluntary adoption across engineering; 45 prototypes produced, 16 adopted into the product and technology roadmap.

VP of AI & Engineering :: Streetbees :: London & Lisbon (Remote) :: *October 2021 - July 2023*

Reports to CPTO. Org of 50+ across 3 stream-aligned teams (acquisition, annotation, reporting).

Series-B AI SaaS – PepsiCo, Unilever, and others use 3.5M+ consumer insights via AI chatbot to understand markets. Owned technical and scientific strategy and operating model; teams ran autonomously.

- **Acquisition:** 75% reduction in human moderation for fraudulent submissions. Algorithm auto-identified 80%+ near-duplicate images and screen captures via a review recommendation engine.
- **Annotation:** Cut manual coding costs 60%+ by replacing a text-only classifier with a cross-modality deep learning model. +300% automated code discovery; removed manual annotation in primary domains (e.g. beauty).
- **Reporting:** Fully automated survey curation, delivery, and report creation using GPT-3. Client delivery: weeks → days. Led to a company pivot and launch of Streetbees-X.

Head of NLP Centre of Excellence :: Zalando :: Berlin (Remote) :: July 2020 - October 2021
Reports to SVP of Applied Science. Single expert team of 10.

Founded and led one of Zalando's four AI Centres of Excellence (NLP, Image Processing, Econometrics, Forecasting). Owned operating model, technical, product, and scientific direction. Incubated capabilities and seeded new product-focused research teams.

- **Platform Capabilities:** GDPR-compliant multilingual feedback search and routing tool. 18 languages, 40 customer touchpoints consolidated into one platform.
- **Content Moderation:** Incubated automated moderation for product safety via [state-of-the-art](#) research. 95%+ recall in flagging unsafe products with human-in-the-loop escalation.
- **Product-Led Scientific Advances:** [Novel text clustering](#) for new feedback and risk categories. Pioneered zero-shot classification on transformers; contributed to [Flair](#) open-source NLP project (Zalando Research).

Supervisory Board Member :: Zalando SE :: May 2018 - June 2020

I served as a member of Zalando's 8-seat supervisory board alongside representatives from Kinnevik, Bestseller Group, Netflix, KPMG, and Rocket Internet, plus two fellow Zalando employee representatives.

Fiduciary oversight covered policy, investments, divestitures, executive promotions and remuneration, plus relationships with strategic investors and key brand partners.

Head of Customer Data Platform :: Zalando :: January 2016 - July 2020
Reports to VP of Engineering. Team of 25 across 3 sub-teams (Preference Recommendations, Price Predictions, Cross-Device Graph).

Founding member of the Customer Data Platform, progressing from Senior Scientist to Head of Applied Science. Cross-functional partnership with Zalando's main business units (Lounge, Markets, Zalon, ZMS, Fashion Store).

- **Customer Preference Recommendations:** Fashion attribute recommendations (category, brand, price) powered preference collection, carousels, personalised navigation, newsletters, and offsite ads. Double-digit A/B test improvements over existing solutions.
- **Audience Targeting:** Predictive audience tools (ZMS, Zalon, Markets) achieved 40x over legacy in early campaigns. Drove client retention externally, deeper market penetration internally.
- **Cross-Device Graph:** Replaced the marketing team's user association model. +40% attributed sales. Integrated into A/B testing platform, reducing test noise by double digits for cookie-based tests.

Senior Applied Scientist :: IBM :: April 2012 - January 2016

Lead Applied Scientist in IBM's SmartCloud division. Invented, optimised, and delivered ML software for IT network monitoring. Started on metrics management; formed a new event analytics team.

- **Event Analytics:** Co-developed from a 2-person discovery team to 14 contributors across USA, UK, Ireland (design, UX, engineering, applied science). 16 co-invented patents. Reduced client workloads 30%, MTTR 10%.
- **Metrics Management:** Redesigned the metrics anomaly product: 1000x I/O improvement (3.5 GB → 3.5 MB), 40x speed-up (5 min → 7 sec), 2.5x memory reduction (70 GB → 20 GB). Doubled detection methods, added memory leak detectors, cut false alerts 50%.

Early Career

Data Scientist :: Swrve :: June 2011 - March 2012 — A/B testing engine and event-stream analytics for mobile gaming

Application Developer :: Murex :: October 2004 - October 2006 — software modules for the Murex Limits Controller (financial risk)

Formal Education:

- University College Dublin, Post Doc, Machine Learning :: Commercialisation; 2010 - 2011
- University College Dublin, Doctor of Philosophy, Machine Learning :: Speaker Verification 2006 - 2010
- Trinity College Dublin, Master of Science :: High Performance Computing, 2002 - 2003
- Trinity College Dublin, B.A. (Mod./Honours) :: Mathematics, 1998 - 2002